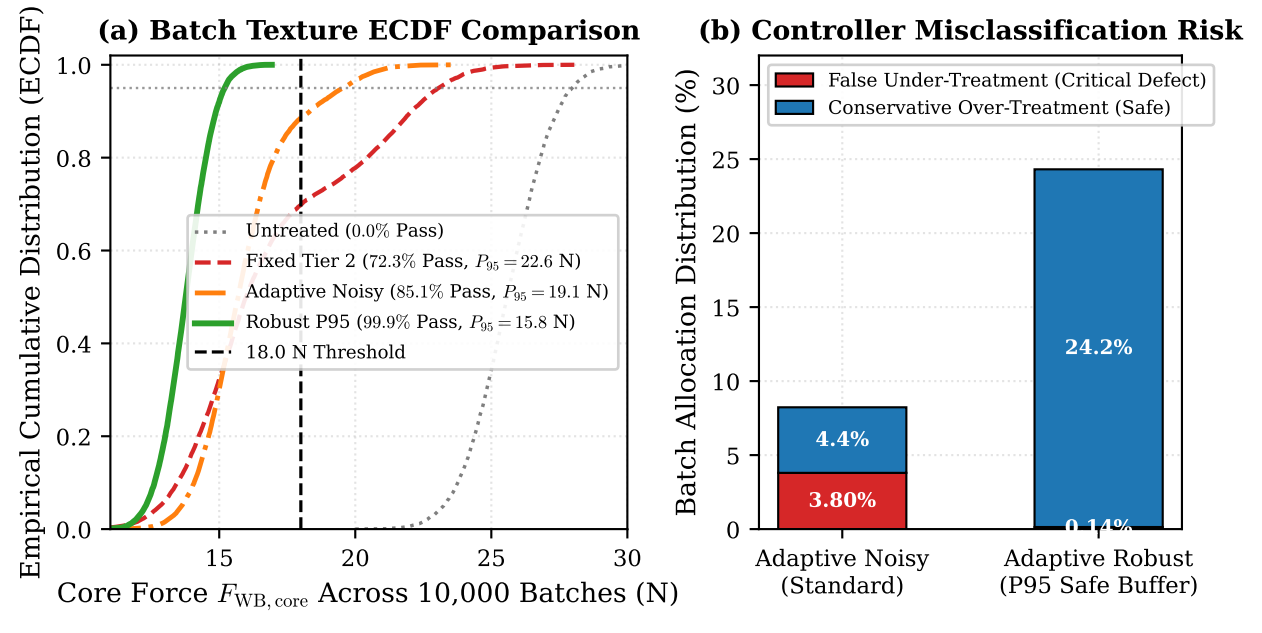

(a) Batch Texture ECDF Comparison
Empirical Cumulative Distribution (ECDF)
Core Force $F_{\mathrm{WB,\,core}}$ Across 10,000 Batches (N)
Untreated (0.0% Pass)
Fixed Tier 2 (72.3% Pass, $P_{95} = 22.6$ N)
Adaptive Noisy (85.1% Pass, $P_{95} = 19.1$ N)
Robust P95 (99.9% Pass, $P_{95} = 15.8$ N)
18.0 N Threshold
(b) Controller Misclassification Risk
Batch Allocation Distribution (%)
False Under-Treatment (Critical Defect)
Conservative Over-Treatment (Safe)
4.4%
3.80%
24.2%
0.14%
Adaptive Noisy (Standard)
Adaptive Robust (P95 Safe Buffer)